

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1102

COURSE OUTCOME COVERED

CO5: Analyze object-oriented software using quality assurance and usability testing Techniques (BL4,BL5)
Assessment Tool Weightage: 35%

QUESTIONS: 5

Total Marks:35

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I Sayed Fazal (192311291) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Constraint-Based Problem Solving Assessment Rubric

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Annexure A

Constraint-Based Problem Solving

Q1. Scenario: Online Banking System

System Components:

Account Management, Fund Transfer, Bill Payment, Transaction History

Given Constraints:

- Transaction completion time < 2 seconds
- Multi-factor authentication required
- Support 5000+ concurrent users

Question:

Analyze the system using quality assurance techniques under the given constraints. Identify potential defects related to security, performance, and reliability. Propose suitable testing strategies and justify how they ensure system quality. **(BL4)**

Answer:

1. Potential Defects

Security Defects

- Weak password or MFA validation.
- Unauthorized access to accounts.
- Session hijacking or improper session timeout.
- Insecure fund-transfer authorization.
- Sensitive transaction data exposed in logs.
- Insufficient protection against SQL injection and other attacks.

Performance Defects

- Transaction response time exceeding 2 seconds.
- Slow database queries for transaction history.
- Performance degradation when many users access the system simultaneously.
- Inefficient fund-transfer or bill-payment processing.

Reliability Defects

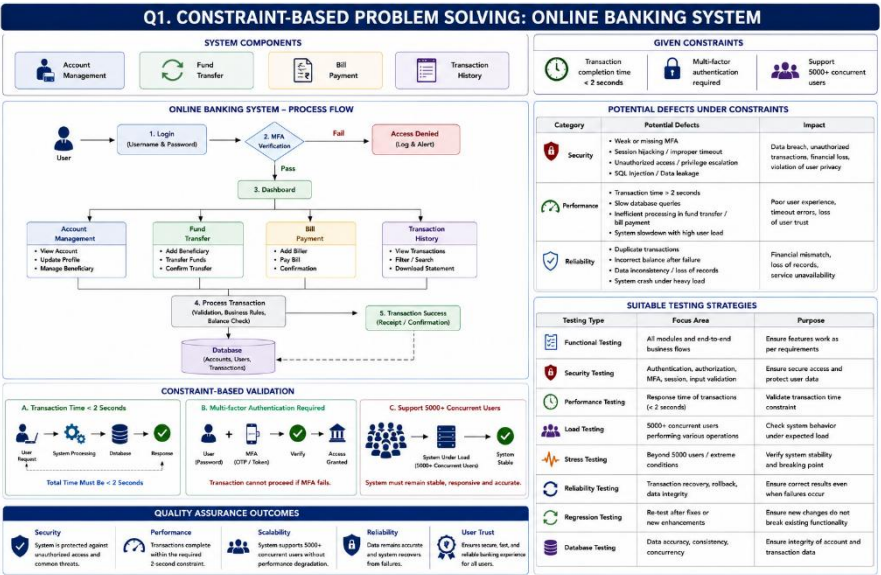
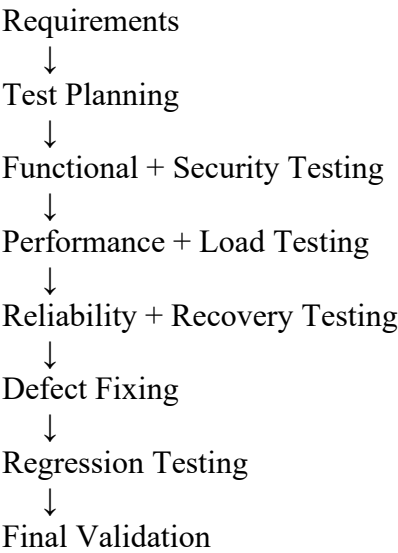
- Duplicate transactions caused by repeated requests.
- Transaction failure without proper rollback.
- Incorrect account balance after failed transfers.
- Loss of transaction records.

- System failure during peak load.

2. Testing Strategies

Testing Strategy	Purpose
Functional Testing	Verifies account management, fund transfers, bill payments, and transaction history work correctly.
Security Testing	Tests MFA, authentication, authorization, session security, and protection against common attacks.
Performance Testing	Measures whether transactions complete within the 2-second constraint .
Load Testing	Tests the system with 5000+ concurrent users .
Stress Testing	Determines system behavior beyond the expected user load.
Reliability Testing	Checks transaction consistency, recovery, rollback, and data integrity after failures.
Regression Testing	Ensures new changes do not break existing banking functions.
Database Testing	Verifies account balances, transaction records, and payment data are stored accurately.

3. Quality Assurance Approach



Q2. Scenario: E-Commerce Shopping Application

System Features:

Product Search, Shopping Cart, Checkout, Payment Gateway

Usability Constraints:

- Checkout process within 3 steps
- Easy product navigation
- Responsive design for mobile and desktop devices

Question:

Apply usability testing techniques to evaluate the application. Identify usability challenges and recommend design improvements that satisfy the given constraints. Explain how the proposed changes improve customer satisfaction and efficiency. **(BL4)**

Answer:

1. Usability Challenges

- **Complex checkout:** Too many forms or unnecessary pages may make checkout lengthy.
- **Difficult navigation:** Poor categories, filters, or search results can make products difficult to find.
- **Unclear interface:** Confusing buttons, icons, or product information can increase user errors.
- **Mobile usability issues:** Fixed layouts or small buttons may not work well on mobile devices.
- **Payment difficulty:** Complicated payment forms can cause users to abandon purchases.
- **Poor feedback:** Users may not know whether an item was added to the cart or payment was successful

Q3. Scenario: Hospital Management System

Modules:

Patient Registration, Appointment Scheduling, Medical Records, Billing

QA Constraints:

- Patient data accuracy must be maintained
- System availability 24/7
- Simultaneous access by multiple departments

Question:

Use quality assurance practices to analyze the system. Identify critical test cases and discuss how the constraints influence testing priorities, defect prevention, and overall system reliability. **(BL5)**

Q4. Scenario: Smart Library Management System

Features:

Book Search, Book Reservation, Borrow/Return, Notifications

Usability Constraints:

- Search results displayed within 2 seconds
- Accessible interface for all age groups
- Minimal learning curve for first-time users

Question:

Conduct a usability evaluation using appropriate testing methods. Analyze how the given constraints impact interface design and recommend improvements to enhance usability and accessibility. **(BL5)**

Q5. Scenario: Airline Reservation System**Modules:**

User Login, Flight Search, Seat Selection, Ticket Booking

QA & Usability Constraints:

- No booking duplication
- System must support peak holiday traffic
- Booking interface should be simple and error-free

Question:

Apply both quality assurance and usability testing techniques under the specified constraints. Identify major risks, propose suitable testing methods, and explain how the system can be optimized for reliability, performance, and user experience. **(BL4)**